JAMES WARNER

Updated October 7, 2026

Research Interests: Atmospheric Dynamics, Convection, Tropical Meteorology, Teleconnections, Numerical Weather Prediction, Process-based Evaluation

PROFESSIONAL ACCREDITATIONS

RMetS
Chartered Meteorologist — *Sep 2023 - Present (3.1 years)*

EMPLOYMENT

UK Met Office — **Total (6.7 years)**

Senior Scientist - Regional Model Evaluation	Mar 2026 - Present (0.6 years)
Manager - Momentum Partnership (Secondment)	Oct 2025 - Mar 2026 (0.4 years)
Senior Scientist - Regional Model Evaluation	Jan 2024 - Oct 2025 (1.8 years)
Scientist - Regional Model Evaluation	Feb 2020 - Jan 2024 (3.9 years)

University of Exeter — **Total (3.4 years)**

Research and Teaching Assistant	Sep 2016 - Feb 2020 (3.4 years)

Other

Tesco: Customer Service Assistant (University Holidays)	May 2013 - Dec 2014 (1.6 years)
Oliver Adams Bakery: Baker/Sales Assistant (Part & Full time)	Mar 2012 - Sep 2012 (0.5 years)
Newport News: Morning Paperboy (Daily inc. Weekends)	Aug 2009 - Sep 2012 (3.3 years)

EDUCATION

University of Exeter — *September 2016 - February 2020*
Ph.D. Mathematics
Thesis: Causality of the link between autumn Arctic sea ice and the winter extratropical atmosphere.

University of Leeds — *September 2015 - September 2016*
M.Res. Climate and Atmospheric Science (Distinction)
Thesis: Understanding changing climatology of extreme precipitation over Europe.

University of Reading — *September 2012 - July 2015*
B.Sc. Meteorology and Climate Science (1st Class Hons.)
Thesis: How do extratropical storm tracks respond to climate change forcings in a simple GCM?

PUBLICATIONS

[17] **Warner, J.** and Flack, D.L., 2026. Modulation of ensemble perturbation growth over tropical Africa by the diurnal cycle. *Quarterly Journal of the Royal Meteorological Society.* doi:10.1002/qj.70316

[16] Bassford, J., Parker, D.J., Marsham, J. and **Warner, J.**, 2025. Scale interactions of deep convection in the Sahel: Exploring ring patterns in lagged cross-correlations of rainfall. *Quarterly Journal of the Royal Meteorological Society*, p.e70041. doi:10.1002/qj.70041

[15] Talib, J., Taylor, C.M., Klein, C., **Warner, J.**, Munday, C., Folwell, S. and Charlton-Perez, C., 2025. Modelling the influence of soil moisture on the Turkana jet. *Quarterly Journal of the Royal Meteorological Society*, p.e4972. doi:10.1002/qj.4972

[14] **Warner, J.L.**, Franklin, C.N., Roux, B., Cooper, S., Rennie, S. and Kumar, V., 2025. Diagnosing lateral boundary spin-up in regional models using an age-of-air diagnostic. *Quarterly Journal of the Royal Meteorological Society*, p.e4971. doi:10.1002/qj.4971

[13] Bush, M., Flack, D.L., Lewis, H.W., Bohnenstengel, S.I., Short, C.J., Franklin, C., Lock, A.P., Best, M., Field, P., McCabe, A., **Warner, J.**, et al., 2024. The third Met Office Unified Model-JULES Regional Atmosphere and Land Configuration, RAL3. *Geoscientific Model Development Discussions*, 2024, pp.1-58. doi:10.5194/gmd-2024-201

[12] Maybee, B., Marsham, J.H., Klein, C.M., Parker, D.J., Barton, E.J., Taylor, C.M., Lewis, H., Sanchez, C., Jones, R.W. and **Warner, J.**, 2024. Wind shear effects in convection permitting models influence MCS rainfall and forcing of tropical circulation. *Geophysical Research Letters*, 51(17), p.e2024GL110119. doi:10.1175/JCLI-D-23-0325.1

[11] **Warner, J. L.** Munday, C., Engelstaedter, S., 2024. Resolving the Turkana Jet: Impact of Model Resolution in Simulating Channel Flow and Inversions. *JGR:Atmospheres*, 129(14), doi:10.1029/2023JD040299

[10] Munday, C., Engelstaedter, S., Washington, R., Ogutu, G., Olago, D., Ouma, G., **Warner, J.**, Ong'ech, D., Nkatha, R., Ogolla, C. and Lees, T., 2024. The Turkana Jet diurnal cycle in observations and reanalysis. *Journal of Climate*, 37(18), pp.4633-4645. doi:10.1175/JCLI-D-23-0325.1

[9] Jones, R.W., Sanchez, C., Lewis, H., **Warner, J.**, Webster, S., Macholl, J., 2023. Impact of domain size on tropical precipitation within explicit convection simulations. *Geophysical Research Letters*, 50(17), doi:10.1029/2023GL104672

[8] **Warner, J.L.**, Screen, J, Scaife, A., Maidens, A., Knight, J., 2023. Tropical forcing of Barents-Kara sea ice during autumn. *Geophysical Research Letters*, 50(8), doi:10.1029/2023GL102768

[7] **Warner, J.L.**, Petch, J., Short, C., Bain, C., 2023. Assessing the impact of an NWP warm-start system on model spin-up over tropical Africa. *QJ*, 149(751), pp.621-636. doi:10.1002/qj.4429

[6] Roberts, B., Clark, A.J., Jirak, I.L., Gallo, B.T., Bain, C., Flack, D.L.A., **Warner, J.**, Schwartz, C.S., Reames, L.J., 2022. Model configuration vs. driving model: influences on next-day regional convection-allowing model forecasts during a real-time experiment. *WAF*, 38(1), pp.99-123. doi:10.1175/WAF-D-21-0211.1

[5] Fletcher, J.K., Diop, C.A., Adefisan, E., Ahiataku, M., Ansah, S.O., Birch, C.E., Burns, H.L., Clarke, S.J., Gacheru, J., James, T.D., Ngetich Tuikong, C.K., Koros, D., Indasi, V.S., Lamptey, B.L., Lawal, K.A., Parker, D.J., Robers, A.J., Stein, T.H.M., Visman, E., **Warner, J.**, Woodhams, B.J., Youds, L.H., Ajayi, V.O., Bosire, E.N., Cafaro, C., Camara, C.A.T., Chanzu, Dione, C., Gitau, W., Groves, D., Groves, J., Hill, P.G., Ishiyaku, I., Klein, C.M., Marhsam, J.H., Mutai, B.K., Ndiaye, P.N., Osei, M., Popoola, T.I., Talib, J., Taylor, C.M., Walker, D., 2022. Tropical Africa's first testbed for high-impact weather forecasting and nowcasting. *BAMS*, 104(8), doi:10.1175/BAMS-D-21-0156.1

[4] Siegert, M., Bacon, S., Barnes, D., Brooks, I., Burgess, H., Cottier, F., Depledge, D., Dodds, K., Edwards, M., Essery, R., Heywood, K., Hendry, K., Jones, V., Lea, J., Medly, I., Meredith, M., Screen, J., Steinberg, P., Tarling, G., **Warner, J.**, Young, G. 2020. The Arctic and the UK: climate, research and engagement. *NORA*. doi:10.25561/80095

[3] **Warner, J.L.**, Screen, J.A., Scaife, A.A., 2019. Links Between Barents-Kara Sea Ice and the Extratropical Atmospheric Circulation Explained by Internal Variability and Tropical Forcing.

Geophysical Research Letters, 47(1), 085679. doi:10.1029/2019GL085679

[2] **Warner, J.L**, 2018. Arctic sea ice; a driver of the winter NAO? *Weather*, 73, pp.307-310. doi:10.1002/wea.3399

[1] Gadian, A.M., Blyth, A.M., Bruyere, C.L., Burton, R.R., Done, J.M., Groves, J., Holland, G., Mobbs, S.D., Pozo, J.T.D., Tye, M.R., **Warner, J.L.**, 2018. A case study of possible future summer convective precipitation over the UK and Europe from a regional climate projection. *International Journal of Climatology*, 38(5), pp.2314-2324. doi:10.1002/joc.5336

PEER REVIEWS COMPLETED

1	Atmospheric Research
3	Climate Dynamics
1	Environmental Research Letters
2	Frontiers in Climate
1	Geophysical Research Letters
4	Journal of Climate
1	Journal of Applied Meteorology and Climatology
1	Journal of Geophysical Research: Atmospheres
1	Nature Communications
3	Quarterly Journal Of Meteorology
1	Weather and Climate Extremes
2	Weather and Forecasting
1	EGU - AMT

SUPERVISION

Sep 2025 -	**Ella T.**	Understanding changing storm dynamics over Africa under climate change and implications for climate risk. (PhD @ Leeds)
May 2025 - Aug 2025	**Maren H.**	Tropical weather feature identification using machine learning (DPhil Rotation Project @ Oxford)
Sep 2022 -	**Kitty A.**	Scale-interaction of convection and its role in key African meteorological processes using novel observations and model simulations over Southern Africa. (DPhil @ Oxford)
Sep 2022 - Sep 2026	**James B.**	Breaking the convection deadlock: scale interactions between deep convection and tropical circulation. (PhD @ Leeds)
Dec 2020 - May 2021	**Jennifer I.**	Classifying high impact events of Africa using machine learning. ("Summer" Placement)

CONFERENCES, WORKSHOPS, TRAINING AND VISITS

Jun 2026	Momentum User Workshop, Met Office (talk)
Mar 2026	Africa MAC + Climate Lab, Oxford
Nov 2025	Climate Lab, Oxford (visit, talk)
Jul 2025	RMetS Atmosphere and Climate Conference, Manchester (poster)
Jun 2025	Momentum User Workshop, UK Met Office (talk(s) and covened session)
May 2025	Africa MAC, Leeds (talk)

Jan 2025	Machine Learning Foundation Course (sessions over 1 month + project)
Dec 2024	Africa MAC, Remote (talk)
Nov 2024	Climate Lab, Oxford (visit)
Jun 2024	Africa MAC, Bristol University (talk)
May 2024	AMS Hurricanes and Tropical Meteorology, LA (talk)
May 2024	NCAR, Boulder (1 week visit)
Mar 2024	CEH Workshop
Feb 2024	Bureau of Meteorology, Melbourne (UNITE 1 month visit)
Nov 2023	SOEE, Leeds (visit)
Oct 2023	WCSSP SA Workshop, UK Met Office (talk)
Oct 2023	Africa MAC, UK Met Office
Aug 2023	Climate Lab, Oxford (visit)
Mar 2023	Africa MAC, Oxford (chaired session, talk)
Feb 2023	SOEE, Leeds (visit)
Jun 2022	UM User Workshop, Remote (talk)
May 2022	Africa MAC, UCL (talk)
May 2022	WCSSP SA Internal Workshop (talk)
Feb 2022	SWIFT Science Meeting, York (talk)
Nov 2021	Africa MAC, Remote (talk)
Jun 2021	WCSSP SA Internal Workshop, Remote (talk)
Jun 2021	UM User Workshop, Remote (talk)
May 2021	AMS Tropical Conference, Remote (talk)
Feb 2020	FORTIS Tropical Meteorology Training Course
Jun 2019	Climate Dynamics Workshop, Exeter University
Jun 2019	UM User Workshop, UK Met Office (poster)
May 2019	AMS Polar Conference, Boulder (talk)
Apr 2019	EGU, Vienna (talk, co-convened session)
Feb 2019	3 month visit to the UK Met Office to run UM Experiments
Jul 2018	RMetS Atmospheric Science Conference, University of York (talk)
Jul 2018	RMetS Student Conference, University of York (talk)
Apr 2018	EGU, Vienna (talk)
Apr 2018	Polar Prediction Training, Abisko (1.5 weeks)
Mar 2018	STEM in Westminster Event (poster)
Mar 2018	NERC GW4+ Wessex Congress, Southampton (talk)
Dec 2017	AGU, New Orleans (poster)
Jul 2017	RMetS Student Conference, UK Met Office (poster + talk)
Apr 2017	NERC GW4+ Annual Confress, Bristol University (poster)
Apr 2017	R+ Statistics Training, Cardiff University (1 week)
Feb 2017	Met Office Academic Partnership Student Event, UK Met Office (poster)
Feb 2017	Arctic and Midlatitude Workshop, Washington DC
Jan 2017	NERC GW4+ Industrial Partners Event, Bristol Science Museum (poster)
Jul 2016	RMetS Student Conference, Manchester University (poster)